A Circuit for Simultaneous Optical Data Reception and Energy Harvesting

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***Abstract*—A circuit for simultaneous optical data reception and energy harvesting using a photovoltaic cell is presented. This circuit is based on a DC-DC boost converter modified to also work as an optical communications receiver. The circuit is able to boost the low voltage of a typical silicon photovoltaic cell to a level that can be used to drive an electronic circuit. The data receiver portion of the circuit comprises a current-sense resistor, two amplification stages and a comparator. Measurements results using a laser diode as the light source show that a bit error rate of 1.25×10^{-3} at 160 kbps is achieved for binary PAM with logic 0 represented by a 5 mW laser beam and a logic 1 with no light. The power delivered by the circuit to a 50 kΩ load is 359.6 μW when the laser beam is modulated at 160 kbps. The conversion efficiency of the converter is 73.4 % and efficiency of the employed solar cell is 15.6 %.**

I. Introduction

Harvesting energy from the environment has the potential of enabling low-power devices to work for extended periods of time, and in some cases indefinitely, without having to be connected to the power grid. Out of all of the ambient sources of energy, light energy is the most abundant and readily available in both indoor and outdoor scenarios with solar energy reaching up to 100 mW/cm^2 density levels in sunny outdoor conditions. Energy harvesting is especially appealing for Internet of Things (IoT) applications where small wireless devices need to work seamlessly without much maintenance in the form of battery replacement or manual battery recharging.

The emergence of high-efficiency light-emitting diode (LED) luminaires has renewed the interest in visible-light optical communications. LEDs can be switched on and off at high speeds unperciptable to the human eye. Hence, LED luminaires can be used for the dual function of providing ambient light and transmitting data [2]–[4]. This dual function of visible light has great potential for sensing applications in indoor settings. Consider, for instance, ambient monitoring in which small sensing units distributed across a building sense temperature, air quality, light levels and occupancy among other ambient variables. Equipped with small solar cells, these sensing units can harvest ambient light energy to power themselves. Typically, the amount of power that these units can harvest from their environment ranges from few micro-watts to few hundreds of micro-watts depending on the size of their photovoltaic cells and the illuminations levels of their environment. Hence, the sensing units must operate with low power budgets which limits their computational speed and demodulation strategies that can be implemented on them.

Figure 1 shows a conceptual diagram of an IoT network composed of small-scale sensing units equipped with a photovoltaic cell for energy harvesting and data reception and with an LED for data transmission. Alternatively, the sensing unit can transmit data using a corner cube retro-reflector or a small scanning mirror [5]. As the figure shows, it is also possible to add a steerable directed light source, such a laser diode, to individually charge and communicate with each sensor node. A directed laser beam has the advantage of being able to provide concentrated light energy for fast charging as well as faster data rates than an LED could. Reusing a photovoltaic solar cell for energy harvesting and data communications has the advantages of lowering fabrication costs and reducing size.

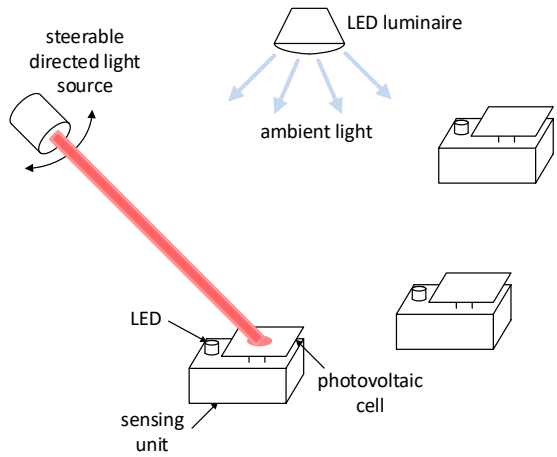

Fig. 1. Conceptual diagram of an IoT network that uses visible light for energy harvesting and optical communications. Each sensing unit in the network is outfitted with a photovoltaic cell (solar cell) for energy harvesting from ambient light and also for data reception.

The concept of employing a solar cell for energy harvesting and data reception has attracted considerable attention recently [4,6]–[10]. In [6] and [7] the authors introduce the concept of employing solar cells (both organic and inorganic) for the dual and simultaneous purpose of energy harvesting and optical communication reception. The work in [6] employed an organic solar cell with an active area of 2 mm × 4 mm making it suitable for small-scale sensing units. A data rate of 34.2 Mbps

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was achieved while harvesting a maximum power of 430 μW when illuminated by a laser diode with output power of 14.3 mW. The work in [7] employed a larger Si solar panel of size 24 cm $\times$ 18 cm and achieved a data rate of 11.8 Mbps while harvesting 30 mW of power from a VESTA 165 mm LED luminaire. Both works employed 4-QAM OFDM with adaptive bit and energy loading. Signal decoding was accomplished using MATLAB and required synchronization, matched filtering, down-sampling, fast Fourier transform, channel estimation, equalization and M-QAM demodulation. Although the power consumption of the receiver was not reported, such a complex receiver will be very challenging to implement on a low-power platform with a power budget of few hundred of micro-watts. In energy harvesting applications where devices have limited computational and energy resources, low-complexity and low-power communication schemes are preferable. In [9] sensing units equipped with a solar cell as a downlink receiver and an LED as an uplink transmitter are proposed. The use of a video camera as the uplink receiver is suggested.

In this work, a circuit for simultaneous optical data reception and energy harvesting using a photovoltaic cell is proposed. This circuit is based on a DC-DC boost converter. The converter is used to boost the low-voltage output (0.3 V to 0.7 V) of silicon photovoltaic cells. In previous work series-connected solar cells have been used to generate large output voltages [7,8]. However, due to the varying nature of ambient light some form of DC-DC conversion and regulation will still be needed in order to provide a stable voltage to the load. Moreover, series-connected solar cells suffer from partial shading effects, which will be manifested if a directed light source, such as a laser, is used to illuminate the cells. To minimize partial shading effects solar cells are connected in parallel and a DC-DC converter is employed to boost their output voltages to levels that match the load power requirements. A proof of concept circuit was built and tested. The light source used was a laser diode with emission centered at 568 nm and 5 mW output power to meet eye safety considerations. Binary Pulse Amplitude Modulation (PAM) was used for modulation of the laser beam output power due to its simplicity which leads to low-complexity demodulation circuits (amplification followed by thresholding) suitable for resource-constrained devices. The output power of the laser diode is varied according to binary data being transmitted.

II. Preliminaries

Figure 2 shows two current-vs-voltage (IV) curves of a typical photovoltaic cell for light intensities that correspond to a transmitted logic 1 and a logic 0. The curves show the characteristic exponential behavior, I_{SC} denotes the short circuit current and V_{OC} the open circuit voltage.

In both curves, the voltage-current points at which the cell generates maximum power, also known as Maximum Power Points or MPPs, are marked. The maximum power generated by the cell when a logic 1 is transmitted is $P_{mpp1} = V_{mpp1} \times I_{mpp1}$ and the maximum power generated when logic 0 is transmitted is $P_{mpp0} = V_{mpp0} \times I_{mpp0}$. Hence, the maximum average power that can be generated by the cell with binary PAM is:

$$P_{max-avg} = p_1 P_{mpp1} + p_0 P_{mmp0} \quad (1)$$

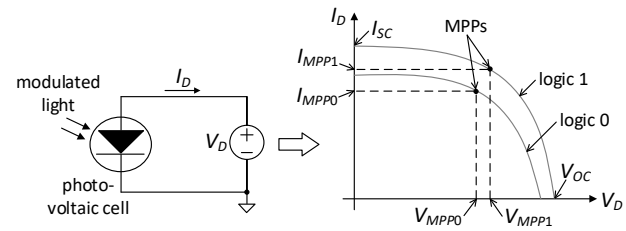

Fig. 2. Current-vs-voltage curves of a typical photovoltaic cell corresponding to a transmitted logic 1 and a logic 0. The Maximum Power Points (MPPs) for both curves are marked.

where, p_1 is the probability of a logic 1 bit being transmitted while p_0 is the probability of a logic 0 bit being transmitted. Typically, $p_1 = p_0 = 0.5$. Hence, $P_{max} = (P_{mpp1} + P_{mpp0})/2$ which is half the power that could be transmitted if the beam is not modulated. Thus, light modulation decreases, on average, the maximum power that can be received with a photovoltaic cell. This power loss can be minimized by decreasing the difference $P_0 - P1$. However, if this difference is made too small the signal-to-noise ratio (SNR) of the received signal will be negatively impacted which in turn will increase the bit error rate (BER). The optimum tradeoff between harvested power and BER will depend on the application priorities. In this work P_0 was set to 5 mW (maximum power regarded as eye safe without protective goggles) and P_1 was set to 0 mW.

Figure 3 shows the circuit employed in [6] and [7] for simultaneous reception of power and data using a photovoltaic cell. The circuit consists of two branches: an energy harvesting branch and a communications branch. Resistor R_L represents the power load while the AC voltage developed across R_C is the communication signal. In this circuit the communication signal is the AC component of voltage V_D. Choke inductor L_0 blocks AC signals from the voltage at the load while capacitor C_0 blocks the DC offset to avoid receiver saturation.

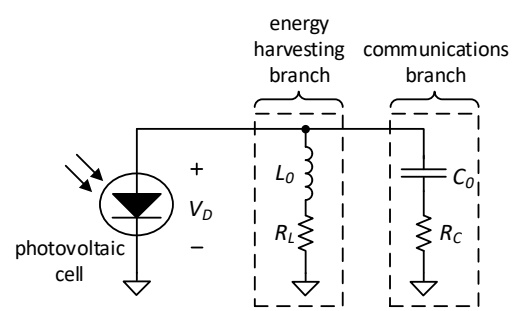

Fig. 3. Schematic diagram of the circuit for simultaneous energy harvesting and communications used in previous works [6] and [7]. The circuit consist of an energy harvesting branch and a communications branch.

A drawback of the circuit in Fig. 3 is that R_L is fixed. However, in order to harvest maximum power from the photovoltaic cell, the load resistor R_L must be varied in response to changes in illumination levels to match the cell's impedance. Moreover, single photovoltaic cells generate low voltages (between 0.3 to 0.7 V depending on illumination conditions) that cannot directly power most electronic circuits. Hence, it is necessary to boost the low voltages generated by the cell. Alternatively

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series-connected solar cells could be used to generate large output voltages [7,8]. However, due to the varying nature of ambient light some form of DC-DC conversion and regulation will still be needed. Moreover, series-connected solar cells suffer from partial shading effects. To counter partial shading, cells are connected in parallel and a DC-DC converter is used to boost their output.

III. DC-DC Converter Circuit for Energy Harvesting and Data Reception

The circuit proposed in this work is based on a switched-inductor DC-DC boost converter. This converter is commonly used in solar energy harvesting and is shown in Fig. 4.

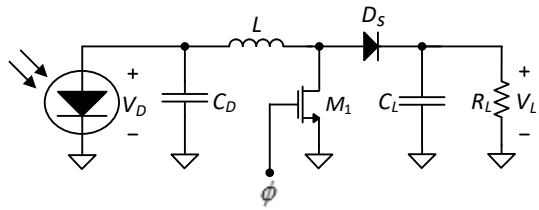

Fig. 4. Schematic diagram of a switched-inductor DC-DC boost converter.

The converter boosts the voltage of the photovoltaic cell, V_D, by transferring packets of energy from the photovoltaic cell to the load R_L through inductor L. This transferring process works in two phases. In the first phase, when ϕ is high, the inductor is charged from the photovoltaic cell and capacitor C_D. In the second phase, when ϕ is low, the energy stored in the inductor is transferred to the load R_L. The diode D_s prevents energy from flowing back from the load to the inductor. Capacitor C_L feeds the load when ϕ is high. If the light intensity varies, the duty cycle of the clock signal ϕ is adjusted to keep maximum power transferred to the load.

This DC-DC boost converter was augmented as shown in Figure 5 to receive binary PAM signals. A current-sense resistor R_s is added in series with the photovoltaic cell to measure the cell's current, I_D as the voltage drop across R_s. This voltage drop is amplified by instrumentation amplifier A_1. Comparator COMP compares the amplifier's output with voltage level V_{CMP} to produce a binary output. A second stage of amplification is achieved with the operational amplifier A_2 and resistors R_1 and R_2.

The switching action of M_1 generates ripple noise on voltage V_C negatively impacting the communication signal. This ripple can be reduced by increasing the value of C_D. However, if C_D is too large, the cell's photocurrent will not be able to charge it in the finite amount of time when ϕ is high. To find the right value od C_D, an experiment in which C_D was varied while the photovoltaic cell was illuminated with a continuous beam with 5 mW of power was performed. The solar cell KXOB22-12X1L was used as the photovoltaic cell, the inductance L was set to 68 μH, $C_L = 100$ μF and $R_L = 50$ kΩ. Different values of C_D were tested while the switching frequency and duty cycle of the clock signal ϕ were varied. For each switching frequency, the duty cycle of ϕ was varied until the maximum load voltage, $V_{L(max)}$, was found. The ripple noise was measured at that duty cycle. Fig. 6 shows $V_{L(max)}$ and the ripple. A value of $C_D = 2.2$ μF and a

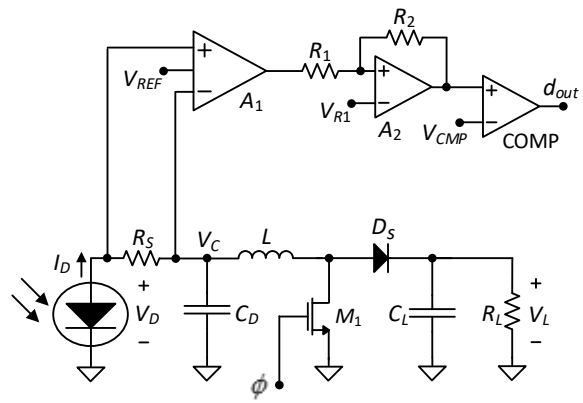

Fig. 5. Schematic diagram of a DC-DC boost converter with additional amplification circuit for data reception.

switching frequency of 40 kHz results in a maximum load voltage of 5.44 V and ripple of 70 mVpp. The ripple noise decreases to 57.5 mVpp for $C_D = 10$ μF but the maximum load voltage drops to 5.35 V.

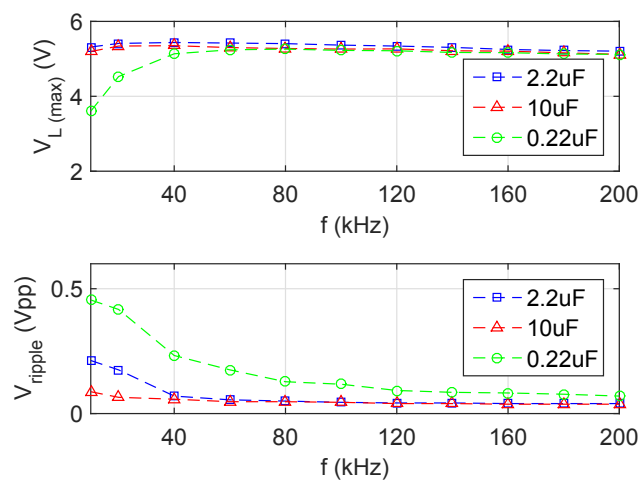

Fig. 6. Maximum voltage at the load and ripple voltage at V_C as a function of the switching frequency.

The power generated by the solar cell was estimated from the cell's IV curve. The IV curve was measured with a source meter (Keithley 2401) while being illuminated with an unmodulated 5 mW light beam. For this measurement the solar cell was disconnected from the rest of the circuit. From the IV curve the maximum power point was found to be $I_D = 2.12$ mA and $V_D = 0.368$ V giving a maximum output power of 780 μW. Hence, the efficiency of the solar cell at 658 nm is 15.6 %. The maximum power delivered to the load for $C_D = 2.2$ μF is $(5.44\ \text{V})^2/50\ \text{k}\Omega = 591.9$ μW resulting in a conversion efficiency for the DC-DC converter of 75.9%. For $C_D = 10$ μF the power delivered to the load is 572.5 μW and the efficiency of the DC-DC converter drops to 73.4%. The lower efficiency is a penalty to pay for the lower ripple noise.

The proposed circuit was also tested as a data receiver. For

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this test, the transmitter circuit shown in Figure 7 was used. A pseudo-random bit sequence was generated by an FPGA and fed to the input V_{in}, after some scaling and inversion so that the power emitted for a 0 bit is 5 mW and the power emitted for a 1 bit is 0 mW. The output of the comparator COMP was fed back to the FPGA and the bit error rate (BER) was calculated for a total of 10^6 transmitted and received bits. The INA321 low-power instrumentation amplifier was used for A_1 and its gain was set to 5 to maximize its bandwidth. This amplifier has a power consumption of 40 μW which makes it suitable for energy harvesting applications. The gain of A_2 was set to 60 by setting $R_1 = 10$ kΩ and $R_2 = 600$ kΩ.

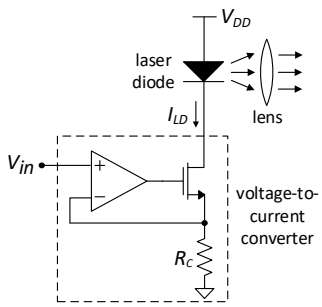

Fig. 7. Schematic diagram of optical transmitter circuit. The voltage-to-current converter formed by the opamp, the transistor and the resistor R_C, generates a current proportional to V_{MOD} which flows through the laser diode. $R_C = 32\ \Omega$.

Table **??** presents the BER results along with the maximum voltage generated at the load. Notably, $C_D = 10\ \mu$F results in best BER performance due to lower ripple noise. At 160 kbps the BER drops to 1.25×10^{-3} but it is still in the limits of forward error correcting codes. Fig. **??** shows the transmitted and received waveforms at 160 kbps. Due to higher ripple noise, the BER increses for $C_D = 2.2\ \mu$F. The limited bandwidth of the solar cell and the low-power amplifiers limits the bit rate for the chosen modulation scheme. Fig. **??** shows the transmitted and received waveforms for a bit rate of 160 kbps.

TABLE I. BER AND $V_{L(max)}$ FOR DIFFERENT BIT RATES

	$C_D = 2.2\ \mu$F		$C_D = 10\ \mu$F	
Bit rate (kbps)	BER	$V_{L(max)}$ (V)	BER	$V_{L(max)}$ (V)
20	0	4.22	0	4.12
40	0	4.23	0	4.16
60	0	4.23	0	4.19
80	0	4.24	0	4.22
100	2.0×10^{-6}	4.29	2.0×10^{-6}	4.22
120	1.4×10^{-4}	4.29	3.0×10^{-6}	4.23
140	9.9×10^{-4}	4.31	1.31×10^{-4}	4.22
150	1.7×10^{-3}	4.33	3.06×10^{-4}	4.22
160	4.2×10^{-3}	4.34	1.25×10^{-3}	4.24

IV. CONCLUSION

A circuit for simultaneous optical data reception and energy harvesting using a photovoltaic cell has been presented. This circuit can be used in sensor network applications where small-scale sensors can harvest energy from ambient light while being able to receive data using a single photovoltaic cell. The proposed circuit is based on a DC-DC boost converter that can also work as a data receiver by the addition of a current-sense resistor, an amplification stage and a comparator. Experimental results show that a bit error rate of 3.06×10^{-4} at 150 kbps and 1.25×10^{-3} at 160 kbps are achieved for binary PAM in which a logic 0 is represented by emitting a 5 mW laser beam and a logic 1 by turning off the laser diode. The power delivered by the circuit to a 50 kΩ load is 359.6 μW when the laser beam is modulated at 160 kbps with binary PAM. An unmodulated beam results in a power of 572.5 μW delivered to the load. The conversion efficiency of the converter was estimated at 73.4 % and efficiency of the solar cell employed was measured to be 15.6 %.

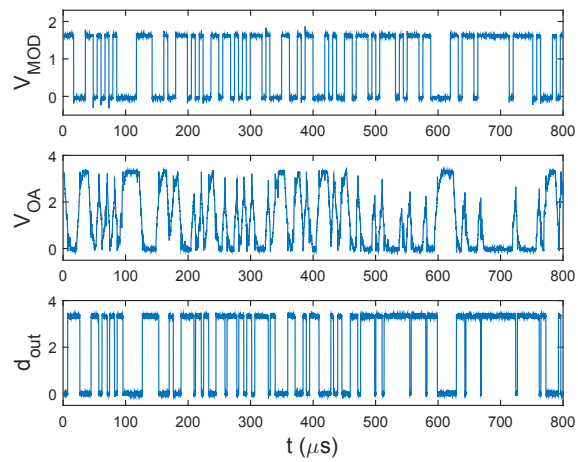

Fig. 8. Acquired waveforms from prototype circuit. V_{MOD} is the modulating signal for the laser diode, V_{OA} is the output of amplifier A_2 and d_{out} is the comparator's output.

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